Universidad de los Andes
Facultad de Economía

Big Data and Machine Learning para Economía Aplicada
MECA 4107
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Problem Set 1: Predicting Income

1 Introduction

In the public sector, accurate reporting of individual income is critical for computing taxes and designing social policy. However, income misreporting remains widespread. According to the Internal Revenue Service (IRS), only about 83.6% of taxes are paid voluntarily and on time in the United States.[1] One of the drivers of this gap is the under-reporting of individual income. Models of labor income therefore play a central role both in understanding inequality and in designing tools for enforcement and targeting.

The objective of this problem set is to develop and evaluate models of individual labor income that could assist the tax authority in detecting potential misreporting. This exercise embodies a fundamental tension in applied econometrics: models that are interpretable and grounded in economic theory may sacrifice predictive accuracy, while flexible predictive models may lack the interpretability needed for policy. This trade-off between interpretability and predictive accuracy is key to the problem set.

The analysis proceeds in three sections. Section 1 estimates an age–income profile motivated by human capital theory. Section 2 examines gender differences in labor income and the choice of controls. Section 3 shifts from interpretation to prediction, comparing specifications based on their out-of-sample performance and examining where they fail. All three sections use data from Bogotá's 2018 household survey.

The analysis is organized around the following question:

What can a model of labor income tell the tax authority about who may be misreporting, and where do its predictions break down?

This question should guide the empirical choices you make throughout the problem set.

2 General Guidelines

2.1 Deliverables

Each team must submit:

- **Slides for in-class presentation.** Prepare **three separate slide decks**, one for each section, and upload each as a `.pdf` to the corresponding activity on Bloque Neón. Use the naming convention `section_equipo_XX.pdf`, where `section` is `age`, `gap`, or `pred`, and `XX` is the two-digit team number. For Team 1:
    - `age_equipo_01` (Section 1: Age–Labor Income Profile)

[1] See https://www.irs.gov/newsroom/the-tax-gap.

   - `gap_equipo_01` (Section 2: Gender Labor Income Gap)
   - `pred_equipo_01` (Section 3: Labor Income Prediction)

- **Public GitHub repository.** Submit a link on Bloque Neón to a **public** GitHub repository containing all code needed to reproduce the analysis and results in the slides. Use the provided template and ensure that each team member makes at least **five (5) substantial contributions** to the main branch. Code should be reproducible, readable, and well-commented, and the README should make the repository easy to navigate. The tidyverse style guide is highly recommended.

2.2 Presentation Format

Each team will deliver a **15-minute in-class presentation**. Teams must be prepared to present **any** of the three sections of the problem set. During class, one group and one presenter will be randomly selected for each section. **All team members must therefore be equally prepared to present all sections.**

Presentations should communicate the empirical analysis clearly to an audience of economists and tell a coherent empirical story. The first slide of each deck is mandatory: a *Result Overview* stating the section's main finding upfront—for example, the estimated peak age and its confidence interval; the unconditional and conditional gender gaps; or the best-performing model and its validation RMSE—before discussing methodology.

Beyond the first slide, teams are free to organize the presentation as they see fit. A recommended flow is to move from the research question and main takeaway, to the descriptive evidence that motivates the analysis, and then to the specifications and results.

Focus on interpretation and conclusions rather than mechanical reporting. Tables and figures should be self-contained, properly formatted, and publication-quality, with clear titles, labeled axes, and legends where appropriate. Tables must not be screenshots from R or other software. Do not include code in the slides, and avoid excessive text.

Golden Rule: Every slide should earn its place: if it does not advance the story the deck is telling, remove it.

General Principle: Every empirical choice in this problem set must be explicitly stated and defensible on economic grounds. Any team member may be asked to justify these choices during the presentation.

2.3 Evaluation Criteria

A full rubric for the presentation and repository can be found in Bloque Neón.

3 Data

3.1 Data Sources and Sample Construction

The data for this problem set are available at https://ignaciomsarmiento.github.io/GEIH2018_sample/. The website hosts microdata for Bogotá from the 2018 *Medición de Pobreza Monetaria y Desigualdad* report, based on the Gran Encuesta Integrada de Hogares (GEIH). The dataset is distributed in **10 separate chunks**, all of which must be scraped. Please be polite when scraping the website and avoid unnecessary requests.

Teams are responsible for cleaning the data and defining a coherent analysis sample to be used throughout the problem set. Unless otherwise stated, restrict the sample to individuals who **report being employed** and are **18 years of age or older**. The main outcome is **total monthly labor income**, which combines income from salaried work and self-employment (`y_total_m`).

3.2 Data Cleaning Decisions

The raw data contain missing values, zero incomes, and potentially implausible observations. There is no single "correct" way to handle these cases. Cleaning decisions should be applied consistently across sections and justified in light of the empirical questions.

3.3 Presenting the Data

Each deck should present only the descriptive evidence that helps answer its section's question. Use descriptive statistics to motivate modeling choices, rather than to document variables or cleaning steps.

4 Section 1: Age–Labor Income Profile

Labor income varies systematically over the life cycle, and human capital theory provides a natural framework for thinking about this relationship. In this section, you will examine the age–income profile in Bogotá and assess how the empirical evidence relates to the predictions of the theory.

Begin by estimating the following *unconditional age–labor income profile*:

$$\log(w) = \beta_1 + \beta_2 \text{Age} + \beta_3 \text{Age}^2 + u,$$

where $\log(w)$ denotes the logarithm of **total monthly labor income**. Use the estimated coefficients to characterize the shape of the age–income profile and determine whether the specification implies a peak within the observed age range. Construct a **bootstrap confidence interval** for the implied peak age.

Next, estimate a *conditional age–labor income profile* by adding total hours worked (`totalHoursWorked`) and employment type (`relab`), and no other controls. Compare the conditional and unconditional profiles and provide an economic interpretation of any differences.

4.1 Minimum Content for the Presentation

The slides must include:

- A regression table comparing the age–income specifications, including the implied peak age, its confidence interval, and a measure of in-sample fit.
- Visualizations of the unconditional and conditional age–income profiles.

Results should be interpreted economically and in relation to human capital theory.

5 Section 2: Gender–Labor Income Gap

This section examines gender differences in labor income and how they change when conditioning on observable characteristics. Unlike Section 1, where the control set was fixed, here the choice of controls is yours to make and defend. This is the central methodological decision in this section.

Begin by estimating the *unconditional gender labor income gap*:

$$\log(w) = \beta_1 + \beta_2 \text{Female} + u$$

Interpret the estimated gender coefficient economically and statistically.

Next, estimate *conditional gender labor income gap* specifications using worker and job characteristics of your choice. Estimate the gender coefficient using the Frisch–Waugh–Lovell (FWL) decomposition and report both analytical and bootstrap standard errors.

5.1 Minimum Content for the Presentation

The slides must include:

- A regression table comparing the unconditional and conditional gender gaps, including analytical and bootstrap standard errors and a measure of in-sample fit.
- Predicted age–income profiles for men and women using a preferred conditional specification, including the implied peak ages and their confidence intervals.

Results should be interpreted economically, with particular attention to how the estimated gender gap changes across specifications. *Be prepared to justify each control you include, discuss whether any may be "bad controls," and explain why the FWL decomposition recovers the same gender coefficient as the full regression.*

6 Section 3: Labor Income Prediction

This section shifts the focus from interpretation to prediction. The objective is to evaluate the out-of-sample performance of the models developed in Sections 1 and 2, improve upon them, and understand what drives the predictions of the best-performing model.

Predictive performance will be evaluated using a **validation set approach**. Use chunks 1–7 as the training sample and reserve chunks 8–10 as the validation sample. Evaluate predictive accuracy using root mean squared error (RMSE) on the validation sample.

Begin by re-estimating the specifications from Sections 1 and 2 on the training sample and evaluating their performance on the validation sample. These models provide the baseline for comparison.

Next, estimate at least **five additional specifications** aimed at improving out-of-sample prediction. Model choices should be economically motivated and may incorporate additional variables, non-linearities, or interactions.

Select the **best-performing model** based on validation-sample RMSE. For this model, compare validation performance with leave-one-out cross-validation (LOOCV) performance on the training sample.[2]

Finally, determine which variable is most important for the predictive performance of your selected model. You must define how you measure variable importance and justify your choice. Once identified, characterize how the model's predictions depend on this variable.

6.1 Minimum Content for the Presentation

The slides must include:

- A table reporting validation-sample RMSE for all estimated specifications, including the models from Sections 1 and 2.
- A comparison of LOOCV and validation-sample performance for the best-performing model.
- An analysis identifying the most important predictor in the selected model, including a clear definition and justification of the variable-importance measure.
- A visualization or other empirical evidence showing how predictions depend on the most important variable.

Results should be interpreted economically and from the perspective of prediction for a tax authority.

[2] A brute-force implementation of LOOCV can be computationally expensive. Think carefully about the most efficient way to implement this calculation using the tools developed in class, and plan accordingly.